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Double Roman domination

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ABSTRACT

For a graph $G = (V, E)$, a double Roman dominating function is a function $f : V \to \{0, 1, 2, 3\}$ having the property that if $f(v) = 0$, then vertex v must have at least two neighbors assigned 2 under f or one neighbor with $f(w) = 3$, and if $f(v) = 1$, then vertex v must have at least one neighbor with $f(w) \geq 2$. The weight of a double Roman dominating function f is the sum $f(V) = \sum_{v\in V} f(v)$, and the minimum weight of a double Roman dominating function on G is the double Roman domination number of G. We initiate the study of double Roman domination and show its relationship to both domination and Roman domination. Finally, we present an upper bound on the double Roman domination number of a connected graph G in terms of the order of G and characterize the graphs attaining this bound.

1. Introduction

The original study of Roman domination was motivated by the defense strategies used to defend the Roman Empire during the reign of Emperor Constantine the Great, 274–337 AD. He decreed that for all cities in the Roman Empire, at most two legions should be stationed. Further, if a location having no legions was attacked, then it must be within the vicinity of at least one city at which two legions were stationed, so that one of the two legions could be sent to defend the attacked city. This part of the history of the Roman Empire gave rise to the mathematical concept of Roman domination, as originally defined and discussed by Stewart [10] in 1999, and ReVelle and Rosing [9] in 2000, and subsequently developed by Cockayne et al. [4] in 2004. Since then about 95 papers have been published on various aspects of Roman domination in graphs, including: (i) Roman domination [1,2], (ii) independent Roman domination [3], and (iii) weak Roman domination [5].

Let $G = (V, E)$ be a graph having *order* $n = |V|$ vertices. The *open neighborhood* of a vertex $v \in V$ is the set $N(v) = \{u|uv \in E\}$, and its *closed neighborhood* is $N[v] = N(v) \cup \{v\}$. Vertices $u \in N(v)$ are called the *neighbors* of v. A vertex with exactly one neighbor is called a *leaf* and its neighbor is a *support vertex*. A support vertex with two or more leaf neighbors is called a *strong support vertex*. The *open neighborhood of a set* $S \subseteq V$ is the set $N(S) = \cup_{v\in S} N(v)$, and the *closed neighborhood of a set* S is the set $N[S] = N(S) \cup S = \cup_{v\in S} N[v]$.

A set $S \subseteq V$ in a graph G is called a *dominating set* if $N[S] = V$. The *domination number* $\gamma(G)$ equals the minimum cardinality of a dominating set in G, and a dominating set of G with cardinality $\gamma(G)$ is called a γ*-set* of G.

Let $f : V \to \{0, 1, 2\}$ be a function having the property that for every vertex $v \in V$ with $f(v) = 0$, there exists a neighbor $u \in N(v)$ with $f(u) = 2$. Such a function is called a *Roman dominating function*. The *weight* of a Roman dominating function

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is the sum $f(V) = \sum_{v \in V} f(v)$. The minimum weight of a Roman dominating function on G is called the *Roman domination number* of G and is denoted $\gamma_R(G)$. A Roman dominating function on G with weight $\gamma_R(G)$ is called a γ_R-*function* of G.

As previously mentioned, the original papers on Roman domination were motivated by the defense strategies used to defend the Roman Empire during the reign of Constantine the Great. But over the years after the time of Constantine, the Roman defense budget significantly exceeded spending at home, and outstripped their ability to raise new revenues. Even at their peak, the Romans could not conquer all of Britain, and so they built Hadrian's Wall, as a defense against the tribes to the north of the wall. But after some time, the Romans finally left all of England. And over the centuries they began a slow, but steady retreat from all conquered lands.

What the Romans badly needed was a more efficient, stronger, yet leaner, defense strategy. In [5], Henning and Hedetniemi proposed weak Roman domination, which provided a reasonable level of defense at a cheaper cost. A vertex v with $f(v) = 0$ is said to be *undefended* with respect to f if it is not adjacent to a vertex w with $f(w) > 0$. A function f is called a *weak Roman dominating function* if each vertex v with $f(v) = 0$ is adjacent to a vertex w with $f(w) > 0$, such that the function $f' = (V_0', V_1', V_2')$ defined by $f'(v) = 1$, $f'(w) = f(w) - 1$, and $f'(u) = f(u)$ for all $u \in V - \{v, w\}$, has no undefended vertex. The *weak Roman domination number* $\gamma_r(G)$ equals the minimum weight of a weak Roman dominating function of G.

Recall that with Roman domination, one legion is required to defend any attacked vertex. What we propose is a stronger version of Roman domination that doubles the protection by ensuring that any attack can be defended by at least two legions. In both Roman domination and weak Roman domination at most two Roman legions are deployed at any one location. But as we will see in what follows, the ability to deploy three legions at a given location provides a level of defense that is both stronger and more flexible, at less than the anticipated additional cost.

A function $f : V \to \{0, 1, 2, 3\}$ is a *double Roman dominating function* on a graph G if the following conditions are met. Let V_i denote the set of vertices assigned i by function f.

(i) If $f(v) = 0$, then vertex v must have at least two neighbors in V_2 or one neighbor in V_3.
(ii) If $f(v) = 1$, then vertex v must have at least one neighbor in $V_2 \cup V_3$.

The *double Roman domination number* $\gamma_{dR}(G)$ equals the minimum weight of a double Roman dominating function on G, and a double Roman dominating function of G with weight $\gamma_{dR}(G)$ is called a γ_{dR}-*function* of G.

One can quickly see a real benefit of double Roman domination. For the *star*, $K_{1,n-1}$, it is easy to see that $\gamma_{dR}(K_{1,n-1}) = 3$ [simply assign the value of 3 to the center vertex of a star]. Note that this is only one more than $\gamma_R(K_{1,n-1}) = 2$. Thus, we can sometimes double the defense of a graph (defend against each attack with at least two legions) with an added cost of no more than 50% of the cost of defending against each attack with one legion.

For another example, consider the complete bipartite graph $K_{2,n-2}$, for $n \geq 4$, where the Roman domination number is 3 (in a partite set of size two, assign the value 2 to one of the vertices and 1 to the other, and assign 0 to all other vertices). But in order to achieve double Roman domination all you have to do is to increase the value of the vertex assigned 1 to 2. Thus, you can double your defenses with an increase in expenditure of only 33%! That is, the Roman domination number equals 3, while the double Roman domination number is only 4.

In Section 2, we consider double Roman domination and its relationship to Roman domination, while in Section 3, we present bounds on the double Roman domination number of a graph G in terms its domination number. Finally, in Section 4, for connected graphs G, we determine a sharp upper bound on $\gamma_{dR}(G)$ in terms of the order of G and characterize the graphs attaining this bound.

2. Double roman domination and roman domination

We begin by showing that for any graph G, there exists a γ_{dR}-function of G where no vertex is assigned a 1, that is, $V_1 = \emptyset$ for some γ_{dR}-function of G.

Proposition 1. *In a double Roman dominating function of weight $\gamma_{dR}(G)$, no vertex needs to be assigned the value 1.*

Proof. Let f be a γ_{dR}-function on a graph G. Suppose that for some $v \in V$, $f(v) = 1$. This means that there is a vertex $u \in N(v)$, such that either $f(u) = 2$ or $f(u) = 3$. If $f(u) = 3$, then we can achieve a double Roman dominating function by reassigning a 0 to v. This results in a function with strictly less weight than f, contradicting that f is a γ_{dR}-function of G. If $f(u) = 2$, then we can create a double Roman domination function g defined as follows: $g(x) = f(x)$ for all $x \notin \{u, v\}$, $g(v) = 0$, and $g(u) = 3$. This results in a double Roman domination function with weight equal to f. □

By Proposition 1, for any double Roman dominating function f', there exists a double Roman dominating function f of no greater weight than f' for which $V_1 = \emptyset$. Henceforth, without loss of generality, in determining the value $\gamma_{dR}(G)$ for any graph G, we can assume that $V_1 = \emptyset$ for all double Roman dominating functions under consideration.

Proposition 2. *Let G be a graph and $f = (V_0, V_1, V_2)$ a γ_R-function of G. Then $\gamma_{dR}(G) \leq 2|V_1| + 3|V_2|$.*

Proof. Let G be a graph and $f = (V_0, V_1, V_2)$ be a γ_R-function of G. We define a function $g = (V_0', V_2', V_3')$ as follows: $V_0' = V_0$, $V_2' = V_1$, and $V_3' = V_2$. Note that under g, every vertex assigned a 0 has a neighbor assigned 3, and no vertex is assigned 1. Hence, g is a double Roman dominating function. Thus, $\gamma_{dR}(G) \leq 2|V_2'| + 3|V_3'| = 2|V_1| + 3|V_2|$, as desired. □

Clearly, the bound of Proposition 2 is sharp, as can be seen with the family of stars $G = K_{1,n-1}$, where $\gamma_R(G) = 2$ and $\gamma_{dR}(G) = 3$. We also note that strict inequality in the bound can be achieved. Consider the subdivided star $G = K^*_{1,k}$, formed by subdividing each edge of the star $K_{1,k}$, for $k \geq 3$, exactly once. We note that $\gamma_R(G) = k + 2$ and $\gamma_{dR}(G) = 2k + 2$. To see this, assign 1 to each leaf, 2 to the center, and 0 otherwise for a Roman dominating function $f = (V_0, V_1, V_2)$; and assign 2 to each leaf, 2 to the center, and 0 otherwise for a double Roman dominating function. It is simple to check that these functions are optimal. Hence, $|V_1| = k$ and $|V_2| = 1$, and so, $2k + 2 = \gamma_{dR}(G) < 2|V_1| + 3|V_2| = 2k + 3$.

Corollary 3. *For any graph G, $\gamma_{dR}(G) \leq 2\gamma_R(G)$ with equality if and only if $G = \overline{K}_n$.*

Proof. Among all γ_R-functions of G, let $f = (V_0, V_1, V_2)$ be one that minimizes the number of vertices in V_1. Since $\gamma_R(G) = |V_1| + 2|V_2|$, by Proposition 2, we have that $\gamma_{dR}(G) \leq 2|V_1| + 3|V_2| = \gamma_R(G) + |V_1| + |V_2| \leq 2\gamma_R(G)$.

If $\gamma_{dR}(G) = 2\gamma_R(G) = 2|V_1| + 4|V_2|$, then since $\gamma_{dR}(G) \leq 2|V_1| + 3|V_2|$, we must have that $V_2 = \emptyset$. Hence, $V_0 = \emptyset$ must hold, and so $V = V_1$. Since $|V_1|$ is minimized under f, we deduce that no two vertices in G are adjacent, for otherwise, if u and v are adjacent, then the function f' which assigns a 0 to u, a 2 to v, and a 1 to every other vertex is a γ_R-function of G having a smaller number of vertices assigned 1 than f does. Thus, $G = \overline{K}_n$. $\square$

From this, the next corollary is immediate.

Corollary 4. *If G is a nontrivial, connected graph, and $f = (V_0, V_1, V_2)$ is a γ_R-function of G that maximizes the number of vertices in V_2, then $\gamma_{dR}(G) \leq 2\gamma_R(G) - |V_2|$.*

By Corollary 4, we see that for a connected graph G, double Roman domination does in fact provide double the protection with strictly less than double the cost of a Roman dominating function. For example, let G be a nontrivial graph with $\gamma(G) = 1$. Then $\gamma_R(G) = 2$ and $\gamma_{dR}(G) = 3$. On the other hand, using the example of the subdivided star $G^* = K^*_{1,k}$, with $k \geq 3$, we see that the $\gamma_{dR}(G)$ approaches $2\gamma_R(G)$ for some graphs. Recall that $\gamma_R(G^*) = k + 2$ and $\gamma_{dR}(G^*) = 2k + 2$, and thus, the ratio of $\gamma_{dR}(G^*)$ to $\gamma_R(G^*)$ is $\frac{2k+2}{k+2}$ and approaches 2 as k approaches infinity.

Next we see that the Roman domination number is strictly smaller than the double Roman domination number.

Proposition 5. *For every graph G, $\gamma_R(G) < \gamma_{dR}(G)$.*

Proof. Let $f = (V_0, V_2, V_3)$ be any γ_{dR}-function of G, where $V_1 = \emptyset$ (by Proposition 1 such a function exists). If $V_3 \neq \emptyset$, then every vertex in V_3 can be reassigned the value 2 and the resulting function will be a Roman dominating function, that is, $\gamma_R(G) < \gamma_{dR}(G)$.

Assume then that $V_3 = \emptyset$. Since $V_2 \cup V_3$ dominates G, it follows that $V_2 \neq \emptyset$. Thus, all vertices are assigned either the value 0 or the value 2, and all vertices in V_0 must have at least two neighbors in V_2. In this case one vertex in V_2 can be reassigned the value 1 and the resulting function will be a Roman dominating function, that is, $\gamma_R(G) < \gamma_{dR}(G)$. $\square$

Corollary 6. *If $f = (V_0, V_2, V_3)$ is any γ_{dR}-function of a graph G, then*

$$\gamma_R(G) \leq 2(|V_2| + |V_3|) = \gamma_{dR}(G) - |V_3|.$$

Corollary 7. *For any nontrivial connected graph G, $\gamma_R(G) < \gamma_{dR}(G) < 2\gamma_R(G)$.*

3. Double roman domination and domination

We first give some straightforward lower and upper bounds on the double Roman domination number in terms of the domination number.

Proposition 8. *For any graph G, $2\gamma(G) \leq \gamma_{dR}(G) \leq 3\gamma(G)$.*

Proof. For the lower bound, let $f = (V_0, V_2, V_3)$ be a γ_{dR}-function of a graph G. Let S be a $\gamma(G)$-set. Note that $(\emptyset, \emptyset, S)$ is a double Roman dominating function. This yields the upper bound of $\gamma_{dR}(G) \leq 3\gamma(G)$.

Further, $V_2 \cup V_3$ is a dominating set for G. Thus, $\gamma(G) \leq |V_2| + |V_3|$. Using this observation, we can obtain the lower bound,

$$\gamma_{dR}(G) = 2|V_2| + 3|V_3| \geq 2(|V_2| + |V_3|) \geq 2\gamma(G). \quad \square$$

Both the bounds of Proposition 8 are sharp. For the upper bound, as we have seen, the family of non-trivial stars $K_{1,n-1}$ has $\gamma(K_{1,n-1}) = 1$ and $\gamma_{dR}(K_{1,n-1}) = 3$. For the lower bound, we also recall that the family of complete bipartite graphs $K_{2,k}$, $k \geq 2$, has $\gamma(K_{2,k}) = 2$ and $\gamma_{dR}(K_{2,k}) = 4$.

The *2-domination number* $\gamma_2(G)$ equals the minimum cardinality of a set S such that every vertex in $V - S$ is adjacent to at least two vertices of S.

Proposition 9. *For any graph G, $2\gamma(G) = \gamma_{dR}(G)$ if and only if $\gamma(G) = \gamma_2(G)$.*

Proof. Let $f = (V_0, V_2, V_3)$ be a γ_{dR}-function on G. Note that $2\gamma(G) \leq 2(|V_2| + |V_3|) \leq 2|V_2| + 3|V_3| = \gamma_{dR}(G)$.

If $2\gamma(G) = \gamma_{dR}(G)$, then both inequalities must be equalities. The first inequality is an equality if and only if $V_2 \cup V_3$ is a minimum dominating set. The second inequality is an equality if and only if $V_3 = \emptyset$. This happens if and only if V_2 is a minimum dominating set and $(V - V_2, V_2, \emptyset)$ is a γ_{dR}-function of G. Hence by definition, every vertex in $V - V_2$ must have at least two neighbors in V_2. Therefore, $\gamma(G) = \gamma_2(G)$. □

As we have seen both the bounds of Proposition 8 are sharp for bipartite graphs. In particular, the lower bound is sharp for the family of complete bipartite graphs $K_{2,k}$ for $k \geq 2$, and the upper bound is sharp for some trees, namely, stars. However, our next result shows that the lower bound can be improved for trees. The *double star* $S_{r,s}$ is the tree with exactly two adjacent non-leaf vertices, one of which is adjacent to r leaves and the other to s leaves. For a tree T rooted at a vertex r and a vertex $v \neq r$ of T, let T_v denote the subtree of T rooted at v, consisting of v and its descendants in T. Further, let $T - T_v$ denote the tree rooted at r formed by removing the subtree T_v from T. Note that the vertices and edges of T_v and the edge from v to its parent in T are removed from T to form $T - T_v$ and that $T - T_v$ is a tree.

Theorem 10. *If T is a non-trivial tree, then $\gamma_{dR}(T) \geq 2\gamma(T) + 1$.*

Proof. Let T be a non-trivial tree of order n. We proceed by induction on n. Note that if T is the star $K_{1,n-1}$, then $\gamma(K_{1,n-1}) = 1$ and $\gamma_{dR}(K_{1,n-1}) = 3 = 2\gamma(K_{1,n-1}) + 1$. Hence, we may assume that $\mathrm{diam}(T) \geq 3$. If $\mathrm{diam}(T) = 3$, then T is the double star $S_{r,s}$ (where $1 \leq r \leq s$), and $\gamma(S_{r,s}) = 2$ and $\gamma_{dR}(S_{r,s}) = 5$ when $r = 1$ and $\gamma_{dR}(S_{r,s}) = 6$ when $r \geq 2$. In either case, $\gamma_{dR}(S_{r,s}) \geq 2\gamma(S_{r,s}) + 1$. Hence, we may assume that the diameter of T is at least 4. This implies that $n \geq 5$.

Assume that any tree T' with order $2 \leq n' < n$ has $\gamma_{dR}(T') \geq 2\gamma(T') + 1$.

We first show that the result holds if T has a strong support vertex. Assume that T has a strong support vertex z adjacent to a leaf u. Since z has at least two leaf neighbors, if any double Roman dominating function of T assigns a value less than 3 to z, the total weight of assigned to z and its leaf neighbors is at least 4. It follows that any γ_{dR}-function of T will assign a 3 to z and 0 to its leaf neighbors. Consider $T' = T - u$. Since z is a support in T', we have that $\gamma(T) = \gamma(T')$ and $\gamma_{dR}(T) \geq \gamma_{dR}(T')$. Moreover, T' is a non-trivial tree, so we can apply our inductive hypothesis showing that $\gamma_{dR}(T) \geq \gamma_{dR}(T') \geq 2\gamma(T') + 1 = 2\gamma(T) + 1$. Since the desired result holds for graphs having a strong support vertex, henceforth we may assume that every support vertex is adjacent to exactly one leaf.

Among all leaves of T, choose r and v to be two leaves such that the distance between r and v is the diameter of T. We root the tree T at vertex r. Let w be the unique neighbor of v, and x be the parent of w. Note that by our choice of v, every child of w is a leaf of T. Since T has no strong support vertices, it follows that $\deg(w) = 2$.

We consider two cases based on the degree of x.

Case 1. $\deg(x) \geq 3$. Let $T' = T - T_w$. Let D be a γ-set of T containing all the support vertices of T (this is possible since every leaf or its support must be in any dominating set), and let D' denote the restriction of D on T'. Now $w \in D$ and $v \notin D$. By our choice of v, every child of x is either a leaf or the support of exactly one leaf. In any case, either x or a child of x is in D. Since $\deg(x) \geq 3$, x is dominated by D'. Thus, $\gamma(T') \leq |D'| = |D| - 1 = \gamma(T) - 1$, that is, $\gamma(T) \geq \gamma(T') + 1$. Conversely, any dominating set of T' can be extended to a dominating set of T by adding w, so $\gamma(T) \leq \gamma(T') + 1$. Thus, $\gamma(T) = \gamma(T') + 1$.

Among all γ_{dR}-functions of T, let f be one such that the weight of f restricted to T_w is minimized. Let f' be the restriction of f onto T'. We claim that f' is a double Roman dominating function of T'. Note that $f(w) \in \{0, 3\}$ and $f(v) \in \{0, 2\}$. Moreover, either $f(w) = 3$ and $f(v) = 0$ or $f(w) = 0$ and $f(v) = 2$.

As we have established, every child of x is a leaf or a support of a leaf. If $f(x) = 0$, then any leaf neighbor of x must be assigned 2 under f, and every support vertex adjacent to x is assigned a 3. If x has two or more support vertices as children, then let g be the function obtained assigning a 2 to x, 0 to each of the children of x that are support vertices, 2 to their leaf neighbors, and $f(a)$ for all vertices in $T - T_x$. If x has three or more support vertices as children, then g is a function with smaller weight than f, contradicting that f is a γ_{dR}-function of T. If exactly two of the children of x are support vertices, then g is a γ_{dR}-function of T with smaller weight restricted to T_w than f has, contradicting our choice of f. Thus, if $f(x) = 0$, we may assume that w is the only child of x that is a support vertex, implying that $\deg(x) = 3$ and x has exactly one leaf neighbor. But then this leaf would be assigned a 2 under f, while $f(w) = 3$. But the function formed from f by assigning $f(v) = 2, f(w) = 0, f(x) = 3$, and 0 to the leaf neighbor of x is a γ_{dR}-function of T, where the weight of T_w is less than under f. Hence, we may assume that $f(x) \geq 2$, that is, f' is a double Roman dominating function of T'. Hence, $\gamma_{dR}(T') \leq \gamma_{dR}(T) - 2$. Now applying our inductive hypothesis, we have $\gamma_{dR}(T) \geq \gamma_{dR}(T') + 2 \geq 2\gamma(T') + 1 + 2 = 2(\gamma(T) - 1) + 3 = 2\gamma(T) + 1$, as desired.

Case 2. $\deg(x) = 2$. Consider $T' = T - T_x$. Since $\mathrm{diam}(T) \geq 4$, T' is a non-trivial tree. Further, since any γ-set of T' can be extended to a dominating set of T by adding the vertex w, we have that $\gamma(T) \leq \gamma(T') + 1$. Note that we can choose a dominating set D of T that does not include x, so $D - \{w\}$ is a dominating set of T'. Hence, $\gamma(T') \leq |D| - 1 = \gamma(T) - 1$, and so, $\gamma(T) = \gamma(T') + 1$.

Among all γ_{dR}-functions of T, let f be one such that the weight of f restricted to T_x is minimized. Let f' be the restriction of f onto T'. Note that $f(w) \in \{0, 3\}$ and $f(v) \in \{0, 2\}$. Moreover, either $f(w) = 3$ and $f(x) = f(v) = 0$ or $f(w) = 0$ and $f(x) = f(v) = 2$. Since the later gives a larger weight when f is restricted to T_x, we will show that this case will not occur.

Suppose to the contrary that $f(x) = f(v) = 2$ and $f(w) = 0$. Then $f(x)$ is necessary to double Roman dominate its parent y (else T_x could have a weight of 3 instead of 4), implying that $f(y) = 0$. Further, y has no neighbor assigned a 3 under f and exactly one other neighbor, say z, assigned 2. But then letting $g(v) = g(x) = 0$, $g(w) = g(z) = 3$, and $g(u) = f(u)$ for all $u \in V - \{v, x, w, z\}$ yields a γ_{dR}-function of T, where the weight assigned to T_x under g is less than the weight under f, a contradiction to our choice of f. Hence, we may assume that $f(w) = 3$ and f' is a double Roman domination function of T'. Thus, $\gamma_{dR}(T') \le \gamma_{dR}(T) - 3$.

Applying our inductive hypothesis, we have that $\gamma_{dR}(T) \ge \gamma_{dR}(T') + 3 \ge 2\gamma(T') + 1 + 3 = 2(\gamma(T) - 1) + 4 = 2\gamma(T) + 2$. □

Proposition 8 and Theorem 10 yield the following for trees.

Corollary 11. *For any tree T, $2\gamma(T) + 1 \le \gamma_{dR}(T) \le 3\gamma(T)$.*

Our final result in this section shows that every value in the range of Corollary 11 is realizable for trees. The *corona* $G \circ K_1$ is the graph formed from G by adding a new vertex v' and edge $v'v$ for each vertex $v \in V(G)$.

Proposition 12. *An ordered pair (a, b) is realizable as the domination number and double Roman domination number of some non-trivial tree if and only if $2a + 1 \le b \le 3a$.*

Proof. Let T be a tree with $\gamma(T) = a$ and $\gamma_{dR}(T) = b$. By Corollary 11, $2a + 1 \le b \le 3a$.

Next we show that each ordered pair is realizable. For $b = 2a + 1$, consider the corona of a star $K_{1,t}$, for $t \ge 1$. It is straightforward to check that $\gamma(K_{1,t} \circ K_1) = t + 1$ and $\gamma_{dR}(K_{1,t} \circ K_1) = 2t + 3 = 2\gamma(K_{1,t} \circ K_1) + 1$.

Now assume that $b \ge 2a + 2$. Let T be the tree formed from a subdivided star $K^*_{1,a}$ by choosing $b - (2a + 2)$ support vertices of $K^*_{1,a}$ and adding another leaf neighbor to each of them. Thus, T has $b - 2a - 2$ strong support vertices. Again, it is straightforward to check that $\gamma(T) = a$ (the set of support vertices form a γ-set of T). To see that $\gamma_{dR}(T) = b$, note that each of the $b - 2a - 2$ strong support vertices must be assigned a 3 under any γ_{dR}-function of T. Assigning a 0 to the support vertices with one leaf, the adjacent leaf a 2 and the center of T a 2. It is simple to check that these functions are in fact minimum. Hence, we have $\gamma_{dR}(T) = 3(b-2a-2)) + 2(a-(b-(2a+2)) + 2 = 3b-6a-6+2a-2b+4a+4+2 = b$, as desired. □

4. Upper bound in terms of order and degree

It is a well known result of Ore [7] that if G is a graph of order n, and has no isolated vertices, then $\gamma(G) \le n/2$. Since we have shown that for any graph G, $\gamma_{dR}(G) \le 3\gamma(G)$, it then follows that for all graphs G of order n, having no isolated vertices, $\gamma_{dR}(G) \le 3n/2$.

In this section we show that for any connected graph G with order $n \ge 3$, $\gamma_{dR}(G) \le 5n/4$. Since deleting an edge cannot decrease the double Roman domination number, it suffices to prove the result for trees. We then characterize the connected graphs attaining the upper bound.

Theorem 13. *If T is a tree with order $n \ge 3$, then $\gamma_{dR}(T) \le 5n/4$.*

Proof. Let T be a tree with order $n \ge 3$. We will proceed by induction on n. Since $n \ge 3$, $\text{diam}(T) \ge 2$. If the diameter of T is 2, then T is the star $K_{1,n-1}$ for $n \ge 3$ and $\gamma_{dR}(T) = 3 < 5n/4$. If the diameter of T is 3, then T is a double star $S_{r,s}$ for $1 \le r \le s$. Hence, $n = r + s + 2 \ge 4$. If $r = 1$, then $\gamma_{dR}(T) = 5 \le 5n/4$. If $r \ge 2$, then $n \ge 6$ and $\gamma_{dR}(T) = 6 < 5n/4$.

Hence, we may assume that the diameter of T is at least 4. This implies that $n \ge 5$.

Assume that any tree T' with order $3 \le n' < n$ has $\gamma_{dR}(T') \le 5n'/4$. Among all longest paths in T, choose P to be one that maximizes the degree of its next-to-last vertex v. Root T at a leaf that is at distance $\text{diam}(T) - 1$ from v. Let u be the parent of v, that is, u is the non-leaf neighbor of v, and let w be a leaf neighbor of v. Note that by our choice of v, every child of v is a leaf. We consider three cases.

Case 1. $\deg(v) > 3$. Then v has a least three leaf children. Consider $T' = T - w$. Therefore, v is a strong support vertex in T', and so v is assigned 3 under any γ_{dR}-function f on T'. Hence, assigning w a 0, we have that f is a double Roman dominating function of T, implying that $\gamma_{dR}(T) \le \gamma_{dR}(T') \le 5n'/4 = 5(n-1)/4 < 5n/4$.

Case 2. $\deg(v) = 3$. Then v has exactly two leaf children. Consider $T' = T - T_v$. Note that since $\text{diam}(T) \ge 4$, T' has at least three vertices, that is, $n' \ge 3$. Let f' be a γ_{dR}-function of T'. Form f from f' by letting $f(x) = f'(x)$ for all $x \in T'$, $f(v) = 3$, and $f(z) = 0$ for all leaf neighbors of v. Then f is a double Roman dominating function of T, implying that $\gamma_{dR}(T) \le \gamma_{dR}(T') + 3 \le 5n'/4 + 3 = 5(n-3)/4 + 3 = (5n-3)/4 < 5n/4$.

Case 3. $\deg(v) = 2$. If $\deg(u) = 2$, then let $T' = T - T_u$. Since $\text{diam}(T) \ge 4$, T' has order $n' \ge 2$. If T' has order 2, then $T = P_5$ and $\gamma_{dR}(P_5) = 6 < 25/4 = 5n/4$. Hence, we may assume that $n' \ge 3$. Applying our inductive hypothesis, we have $\gamma_{dR}(T') \le 5n'/4$. Form f from any γ_{dR}-function of T' by assigning 3 to v, and 0 to each of u and w. Then $\gamma_{dR}(T) \le \gamma_{dR}(T') + 3 \le 5n'/4 + 3 = 5(n-3)/4 < 5n/4$.

Hence, we may assume that $\deg(u) \ge 3$. By our choice of v, every child of u in T_u is either a leaf or a vertex similar to v, that is, a support vertex adjacent to exactly one leaf. Let $T' = T - T_v$. Then $n' \ge 3$ and our inductive hypothesis applies to T'. If there exists a γ_{dR}-function f' of T' such that $f'(u) \ge 2$, then assigning 0 to v and 2 to w and the value $f'(x)$ to all other vertices yields a double Roman dominating function of T. Hence, $\gamma_{dR}(T) \le \gamma_{dR}(T')+2 \le 5n'/4+2 = 5(n-2)/4+2 < 5n/4$.

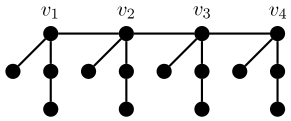

Fig. 1. A tree $T \in \mathcal{F}$.

Henceforth, we may assume that no γ_{dR}-function of T' assigns at least a 2 to u, that is, u must be assigned a 0 or a 1 by every γ_{dR}-function of T'. By Proposition 1, we may assume that u is assigned a 0 under some γ_{dR}-function f' of T'. Let r be the number of children of u, other than v, with degree 2 and s be the number of leaf children of u in T'. If $s \geq 2$, then u would be assigned a 3 under any γ_{dR}-function of T', a contradiction. Similarly, if $r \geq 2$, then u could be assigned at least 2 under some γ_{dR}-function of T'. Hence, $r \leq 1$ and $s \leq 1$. If $r + s = 2$, then again u could be assigned a 3 under some γ_{dR}-function of T', namely, a function that assigns 3 to u, 0 to the two children of u in T' and 2 to the leaf neighbor of a child of u in T'', a contradiction. Hence, $r + s \leq 1$.

Assume that $r = 1$ and $s = 0$. Consider $T'' = T - T_u$. Since the $\deg(u) \geq 3$ and $s = 0$, T'' has order $n'' \geq 2$. If T'' has order 2, then T is the subdivided star $K^*_{1,3}$. Thus, $\gamma_{dR}(T) = 8 < 35/4 = 5n/4$. Hence, we may assume that $n'' \geq 3$. Applying our inductive hypothesis, we have $\gamma_{dR}(T'') \leq 5n''/4$. Form f from any γ_{dR}-function of T'' by assigning 2 to u, 0 to the two children of u and 2 to each of their leaf neighbors. Then $\gamma_{dR}(T) \leq \gamma_{dR}(T'') + 6 \leq 5n''/4 + 6 = 5(n-5)/4 + 6 = (5n-1)/4 < 5n/4$.

Finally, assume that $r = 0$ and $s = 1$. Again, we let $T'' = T - T_u$. Since the diameter of T is at least 4, T'' has order $n'' \geq 2$. If T'' has order 2, then T is the corona of the star $K_{1,2}$. Thus, $\gamma_{dR}(T) = 7 < 30/4 = 5n/4$. Hence, we may assume that $n'' \geq 3$. Applying our inductive hypothesis, we have $\gamma_{dR}(T'') \leq 5n''/4$. Form f from any γ_{dR}-function of T'' by assigning 3 to u, 0 to the two children of u and 2 to the leaf neighbor of the child of u with degree 2. Then $\gamma_{dR}(T) \leq \gamma_{dR}(T'') + 5 \leq 5n''/4 + 5 = 5(n-4)/4 + 5 = 5n/4$. $\square$

Corollary 14. *If G is a connected graph with order $n \geq 3$, then $\gamma_{dR}(G) \leq 5n/4$.*

We note that the bound of Theorem 13 is sharp and characterize the trees T attaining the bound. For this purpose, we define the family $\mathcal{F}$ of trees as follows:

Let T be a tree and let v be a non-leaf vertex of T. Define $f(T, v)$ be a tree obtained from T by adding a P_4 with a support vertex u to T via the edge uv. Let $\mathcal{F}$ be the smallest family of graphs such that: *(i)* $\mathcal{F}$ contains P_4, and *(ii)* if $T \in \mathcal{F}$ and v is a non-leaf vertex of T, then $f(T, v) \in \mathcal{F}$. In other words, a tree T in $\mathcal{F}$ can be built from k copies of P_4 by adding $k - 1$ edges incident to support vertices of the kP_4 to connect the graph. See Fig. 1 for an example.

Theorem 15. *The family $\mathcal{F}$ is precisely the family of trees for which $\gamma_{dR}(T) = 5n/4$.*

Proof. Let T be a tree in $\mathcal{F}$. Observe that if H is an induced subgraph of T such that H is isomorphic to P_4 and the leaves of H are leaves of T, then every double Roman dominating function of T assigns a weight of at least 5 to H. By our construction of T, T has $n/4$ disjoint copies of the induced subgraph P_4, where the leaves of the P_4 are leaves of T. Hence, $\gamma_{dR}(T) \geq 5(n/4) = 5n/4$. By Theorem 13, $\gamma_{dR}(T) = 5n/4$.

To show that every tree T that has $\gamma_{dR}(T) = 5n/4$ is in $\mathcal{F}$, we appeal to our proof of Theorem 13. The proof is by induction on n. Since $P_4 \in \mathcal{F}$, and n is a multiple of 4, we may assume that $n \geq 8$. From the proof of Theorem 13, there is only one case, namely, a subcase of Case 3, where it is possible to achieve equality. Using the terminology from this proof, we have that T is rooted at a peripheral vertex, the diameter of T is at least 4, and v is a vertex adjacent to a vertex that is maximum distance from the root. Further, u is the parent of v, that is, u is the non-leaf neighbor of v, and let w is a leaf neighbor of v. Recall that $T'' = T - T_u$. In the case where equality is achieved T_u is isomorphic to P_4 and the leaves of T_u are leaves of T. Moreover, $5n/4 = \gamma_{dR}(T) \leq \gamma_{dR}(T'') + 5 \leq 5n''/4 + 5 = 5(n-4)/4 + 5 = 5n/4$, implying equality throughout. Thus, $\gamma_{dR}(T'') = 5n''/4$, and so, $T'' \in \mathcal{F}$. Now clearly, $T \in \mathcal{F}$, since $f(T'', x) = T$, where x is the parent of u in T. $\square$

Let $\mathcal{H}$ be the family of connected graphs G of order n that can be built from $n/4$ copies of P_4 by adding a connected subgraph on the set of centers of $\frac{n}{4}P_4$.

Theorem 16. *The family $\mathcal{H}$ is precisely the family of connected graphs G for which $\gamma_{dR}(G) = 5n/4$.*

Proof. Let $G \in \mathcal{H}$. As in the proof of Theorem 15, we have that $\gamma_{dR}(G) = 5n/4$.

Now assume that $\gamma_{dR}(G) = 5n/4$. Since removing edges cannot decrease $\gamma_{dR}(G)$, it follows that every spanning tree of G has double Roman domination number $5n/4$. Thus, every spanning tree of G is in $\mathcal{F}$. If any edge joins the leaves of a spanning tree, then a double Roman domination function with a smaller weight can be assigned. Hence, $G \in \mathcal{H}$. $\square$

5. Summary and open problems

In the preceding sections we have introduced a new model of Roman domination, double Roman domination, which is designed to offer twice the degree of protection provided by Roman domination, but at less than twice the additional cost, in most cases. The following inequalities illustrate some of the relationships between the domination, Roman domination and double Roman domination numbers:

$$\gamma_R(G) \le \gamma_{dR}(G) \le 2\gamma_R(G).$$
$$\gamma(G) \le \gamma_R(G) \le 2\gamma(G) \le \gamma_{dR}(G) \le 3\gamma(G).$$

Graphs having $\gamma_R(G) = 2\gamma(G)$ are called *Roman Graphs* in [4]. Similarly, we say that a graph G is a *double Roman Graph* if $\gamma_{dR}(G) = 3\gamma(G)$. An open problem is to characterize the double Roman graphs. In particular, characterize the double Roman trees.

A theorem of McQuaig and Shepherd [6] proves, that with the exception of seven graphs, six of order seven, and the four cycle C_4, every connected graph G having minimum degree at least two satisfies the inequality, $\gamma(G) \le 2n/5$. Since $\gamma_{dR}(G) \le 3\gamma(G)$, this implies that with these seven exceptions, every connected graph G having minimum degree at least two satisfies the inequality, $\gamma_{dR}(G) \le 6n/5$. Can this bound be improved?

Similarly, a theorem of Reed [8] proves that every connected graph G having minimum degree at least three satisfies the inequality, $\gamma(G) \le 3n/8$. From this we can conclude that these graphs also satisfy the inequality, $\gamma_{dR}(G) \le 9n/8$. Can this bound be improved?

These two observations and Theorem 13 raise the question: for which classes of graphs, or trees, is $\gamma_{dR}(G) \le n$?

We have shown that $2\gamma(G) = \gamma_{dR}(G)$ if and only if $\gamma(G) = \gamma_2(G)$. Another problem is to characterize the class of graphs for which $\gamma(G) = \gamma_2(G)$. This class includes the complete multipartite graphs $K_{2,n_1,n_2,\ldots,n_k}$ where $2 \le n_1 \le n_2 \le \cdots \le n_k$.

By Proposition 1, for any double Roman dominating function f', there exists a double Roman dominating function f of no greater weight than f' for which $V_1 = \emptyset$. Throughout this paper in determining the value $\gamma_{dR}(G)$ for any graph G, we assumed that $V_1 = \emptyset$ for the double Roman dominating functions under consideration. This assumption does not affect the weight of a minimum double Roman dominating function but it could affect the number of vertices in $V_1 \cup V_2 \cup V_3$. Considering defensive strategies, in some cases it is desirable to maximize the number of locations where legions are stationed, that is, to maximize $|V_1 \cup V_2 \cup V_3|$. Indeed, although we do not pursue this question in this paper, it might be worthwhile to seek a γ_{dR}-function in which a maximum number of vertices are in $V_1 \cup V_2 \cup V_3$.

Finally, is it possible to construct a polynomial algorithm for computing the value of $\gamma_{dR}(T)$ for any tree T?

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